WTD2-P SERIES · PERMANENT MAGNET GEARLESS

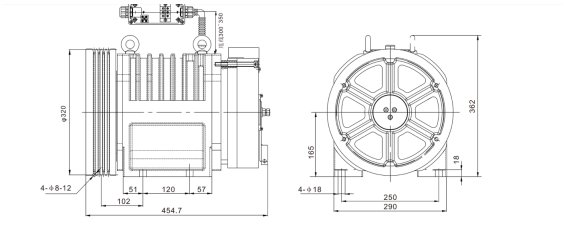
WTD2-P400-100D

400 / 408 kg · 1.0 m/s

Compact permanent magnet synchronous machine for residential and small commercial elevators.

REFERENCE UNIT · WTD2-P400

DIMENSION DRAWING · MM



ELECTRICAL CHARACTERISTICS

Rated power	**2.7 kW**
Rated current	**8 A**
Rated speed	**119 rpm**
Frequency	**19.9 Hz**
Rated torque	**217 N·m**
Number of poles	**20**
Rated voltage	**AC 380 V · 50/60 Hz**
Brake voltage	**DC 110 V · single coil**

MECHANICAL PROPERTIES

Net weight	**140 kg**
Total length	**454.7 mm**
Overall height	**362 mm**
Mounting height	**165 mm**
Mounting holes	**4 × Ø18 @ 250 mm**
Sheave diameter	**Ø 320 mm**
Static shaft load	**2,500 kg**
Travelling height	**≤ 90 m**

DESIGN & STANDARDS

Motor type **PM synchronous**	Wrap **Single**	Roping **4 × Ø8 × 12 · ratio 2:1**
Groove profile **Undercut U**	Cut angle β **90°**	Groove angle γ **30°**
Duty cycle **S5 – 40%**	Insulation class **F**	Protection **IP41**

STANDARDS & CERTIFICATIONS

EN 81	**CE**	**IEC 60034**	**Class F**	**IP41**
European elevator safety	Compliant configurations	Rotating electrical machines	Insulation rating	Ingress protection

NOTES

- Specifications stated at rated voltage and rated load. Operating conditions outside the rated envelope may shift performance.
- Drawings dimensioned in millimetres. Tolerances ±0.5 mm on mounting hole spacing. Refer to engineering for full GA drawings.
- Configurations beyond the published baseline (alternative voltages, groove geometries, sheave diameters) available on request.

© ASTON. DS-WTD2-P400-100D · Rev 1.0 · April 2026. Specifications subject to change without notice. Issued for technical evaluation.